8.8 Period Three Implies Chaos

WORKSHEET

The question

Let $\mathrm{f}(x) = 7 - 2|x|$. A sequence $u_0, u_1, u_2, \ldots$ is defined by $u_0 = a$ and $u_n = \mathrm{f}(u_{n-1})$ for $n > 0$.

(i) (a) Sketch, on the same axes, the graphs with equations $y = \mathrm{f}(x)$ and $y = \mathrm{f}(\mathrm{f}(x))$.

(b) Find all solutions of the equation $\mathrm{f}(\mathrm{f}(x)) = x$.

(c) Find the values of a for which the sequence $u_0, u_1, u_2, \ldots$ has period 2.

(d) Show that, if $a = \frac{28}{5}$, then the sequence $u_2, u_3, u_4, \ldots$ has period 2, but neither u_0 nor u_1 is equal to either of u_2 or u_3.

(ii) (a) Sketch, on the same axes, the graphs with equations $y = \mathrm{f}(x)$ and $y = \mathrm{f}(\mathrm{f}(\mathrm{f}(x)))$.

(b) Consider the sequence $u_0, u_1, u_2, \ldots$ in the cases $a = 1$ and $a = -\frac{7}{9}$. Hence find all the solutions of the equation $\mathrm{f}(\mathrm{f}(\mathrm{f}(x))) = x$.

(c) Find a value of a such that the sequence $u_3, u_4, u_5, \ldots$ has period 3, but where none of u_0, u_1 or u_2 is equal to any of u_3, u_4 or u_5.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself What does the graph of $y = 7 - 2|x|$ look like? Where does it cross the line $y = x$? Where does it send the interval $[-7, 7]$?

STAGE 1 The graph of $f \circ f$

Ask yourself As x goes from -7 to 7, $f(x)$ goes from -7 up to 7 and back down to -7. What does $f(f(x))$ do?

STAGE 2 Solving $f(f(x)) = x$

Ask yourself Each of the four straight pieces crosses the diagonal $y = x$ exactly once. Two of the crossings you already know. Why? How can you find the other two without expanding nested absolute values?

STAGE 3 Arriving at a cycle late

Ask yourself How could you have found $\frac{28}{5}$ yourself? Work backwards: which numbers does f send to $-\frac{7}{5}$?

STAGE 4 Three-cycles

Pause and try Compute the sequences starting from $a = 1$ and $a = -\frac{7}{9}$.

Ask yourself How many solutions can $f(f(f(x))) = x$ have? Think about the shape of its graph, as in Stage 1.

Try this next

1. **A 4-cycle.** Use the covering diagram to predict where a 4-cycle of f should be (three steps in I_2, one in I_1). Then find it exactly. (*Hint:* the cycle has three positive points and one negative point; write down the four linear equations.)

2. **The subtlety.** In the proof that period 3 implies period n, explain why the point x found cannot be the point 1 (or any point of the original 3-cycle) when $n \geq 4$.

3 **Period 5 without period 3.** Sketch the graph of a continuous function on $[1,5]$ with $g(1)=3$, $g(3)=4$, $g(4)=2$, $g(2)=5$, $g(5)=1$, joining these points with straight lines. Draw its covering diagram for the intervals $[1,2]$, $[2,3]$, $[3,4]$, $[4,5]$. Can you find a loop of length 3? (Sharkovskii's theorem says there is no 3-cycle.)

4 **The tent map.** Check that $x = 14y - 7$ turns f into the tent map. Then show that the tent map doubles y written in binary, flipping all the digits whenever the leading digit is 1, and use this to find its 3-cycles.

Further reading

- **Tien-Yien Li & James Yorke, "Period three implies chaos"**, *American Mathematical Monthly* 82 (1975). Short, and readable with the intermediate value theorem.
- **Robert Devaney, *A First Course in Chaotic Dynamical Systems*** (CRC Press). Written for undergraduates; covers the tent map, the logistic map and Sharkovskii's theorem with full proofs.
- **Keith Burns & Boris Hasselblatt, "The Sharkovsky theorem: a natural direct proof"**, *American Mathematical Monthly* 118 (2011). A modern proof using covering diagrams.

The full guided solution, and what the question leads to at university, is in Section 8.8:
https://beyondstep.keithmcnulty.org/ch08/period-three.html